

Reader: I certainly prefer to be given the foundation.

Author: Our next dialogue will be devoted to a discussion of the programme of actions as outlined above. At the first step of the programme, the main rules for differentiation will be established on the basis of the relations of (4), and, in addition, the derivatives of three functions: $y=x$, $y=\sin x$, and $y=\log x$ will be obtained. At the second step, we shall obtain (without reference to the relations of type (4)) the derivatives of the following functions: $y=x^n$, $y=x^{-n}$, $y=\sqrt{x}$, $y=\cos x$, $y=\tan x$, $y=\cot x$, $y=\arcsin x$, $y=\arccos x$, $y=\arctan x$, $y=\operatorname{arccot} x$ and $y=a^x$.

Reader: I will be looking forward to the next dialogue. By the way, you wanted to make one more remark about the derivative $f'(x)$.

Author: [Note 2] concerns the natural domain of a derivative. Let a set D be the domain of $f(x)$. The question is whether D is also the domain of $f'(x)$.

Reader: In any case, the domain of $f'(x)$ cannot be wider than the domain of $f(x)$ because in order to find $f'(x)$ we use $f(x)$.

Author: A carefully balanced answer, to be sure. The domain of $f'(x)$ is in the general case a subset of D . It is obtained from D as a result of elimination of those points x for which $\lim_{\Delta x \rightarrow 0} \frac{\Delta f(x)}{\Delta x}$ does not exist. By the way, this subset is called the domain of differentiability of $f(x)$.

Reader: What are the conditions of differentiability of $f(x)$ at any specific point x ?

Author: Obviously, these conditions are identical to those of the existence of $\lim_{\Delta x \rightarrow 0} \frac{\Delta f(x)}{\Delta x}$ at point x . We have already observed that it is the limit of the type $\frac{0}{0}$, which necessitates that both the numerator and denominator tend to zero. It means that $f(x)$